CS1237 User Manual

24-bit Sigma-Delta ADC

Rev 1.1

CHIPSEA (Shenzhen Chipsea Technologies)

English translation of the original Chinese datasheet. Figures are described in text or redrawn schematically; refer to the original for exact drawings. In case of doubt the Chinese original prevails.

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Revision History

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REV 1.0	Initial version	2014-3-26
REV 1.1	1. Format corrections 2. Differential input impedance parameter changed 3. P-P noise parameter changed 4. Common-mode input range parameter changed	2014-10-17

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1 Chip Description

The CS1237 is a high-precision, low-power analog-to-digital converter with one differential input channel, a built-in temperature sensor and a high-accuracy oscillator.

Selectable PGA gain: 1, 2, 64, 128 (default 128).

Selectable ADC output data rate in normal mode: 10 Hz, 40 Hz, 640 Hz, 1.28 kHz (default 10 Hz).

An MCU communicates with the CS1237 over a 2-wire SPI interface (SCLK and DRDY/DOUT) to configure it, e.g. channel selection, PGA selection, output data rate selection.

1.1 Main Features

- Built-in oscillator
- Integrated temperature sensor
- Power-down function
- 2-wire SPI interface, up to 1.1 MHz

ADC features:

- 24-bit no-missing-codes
- Selectable PGA gain: 1, 2, 64, 128
- 1 differential input, 24-bit no-missing-codes; at PGA = 128 ENOB is 20 bits (5 V) / 19.5 bits (3.3 V)
- P-P noise: PGA = 128, 10 Hz: 180 nV
- INL below 0.0015 %
- Selectable output rate: 10 Hz, 40 Hz, 640 Hz, 1.28 kHz
- Internal short function

1.2 Applications

- Industrial process control
- Electronic scales
- Liquid / gas chemical analysis
- Blood glucose / blood analysis meters
- Smart transmitters
- Portable devices

1.3 Basic Architecture

The CS1237 is a high-precision, low-power Sigma-Delta ADC with one built-in Sigma-Delta ADC, one differential input channel and one temperature sensor. The ADC uses a 2nd-order sigma-delta modulator. The PGA is implemented with a low-noise instrumentation amplifier structure, with selectable gain of 1, 2, 64 or 128. At PGA = 128 the effective resolution reaches 20 bits (at 5 V supply).

The CS1237 has an internal RC oscillator; no external crystal is needed.

Via DRDY/DOUT and SCLK the CS1237 can be configured for several functional modes, e.g. temperature measurement, PGA selection, ADC output rate selection.

The CS1237 has a Power-down mode.

Figure 1 – Block diagram (text description): pins AINP/AINN and the temperature sensor feed an input multiplexer → PGA (1/2/64/128) → ΔΣ ADC. The ADC is clocked by the internal oscillator and uses REFIN/REFOUT as reference. Digital interface: DRDY/DOUT and SCLK. Supply: VDD, GND.

1.4 Absolute Maximum Ratings

Table 1 – CS1237 absolute maximum ratings

Name	Symbol	Min	Max	Unit
Supply voltage	VDD	-0.3	6	V

Name	Symbol	Min	Max	Unit
Supply surge current			100	mA
Supply continuous current			10	mA
Digital pin input voltage		-0.3	DVDD+0.3	V
Digital output pin voltage		-0.3	DVDD+0.3	V
Junction temperature			150	°C
Operating temperature		-40	85	°C
Storage temperature		-60	150	°C
Lead soldering temperature			240	°C

1.5 Digital Logic Characteristics

Table 2 – CS1237 digital logic characteristics

Parameter	Min	Typ	Max	Unit	Condition
VIH	0.7×DVDD		DVDD+0.1	V	
VIL	DGND		0.3×DVDD	V	
VOH	DVDD-0.4		DVDD	V	Ioh = 1 mA
VOL	DGND		0.2×DVDD	V	IoL = 1 mA
IIH			10	μA	VI = DVDD
IIL	-10			μA	VI = DGND
Serial clock SCLK frequency			1.1	MHz	

1.6 Electrical Characteristics

All parameters are tested at ambient temperature -40 to 85 °C with the internal reference, unless otherwise noted.

Table 3 – CS1237 electrical characteristics (VDD = 5 V, 3.3 V)

Parameter	Condition	Min	Typ	Max	Unit
Analog input					
Full-scale input voltage (AINP-AINN)			±0.5·VREF/PGA		V
Common-mode input voltage	PGA = 1, 2	AGND-0.1		AVDD+0.1	V
	PGA = 64, 128	AGND+0.75		AVDD-0.75	V
Differential input impedance	PGA = 1, 2		190		MΩ
	PGA = 64, 128		28		MΩ
System performance					
Resolution	No missing codes		24		Bits
ADC data rate			10	1280	Hz
Settling time	Fully settled		3: output rate 10/40 Hz 4: output rate 640/1280 Hz		Conversion cycles
P-P noise	PGA = 128, 10 Hz		180		nV
Effective resolution	PGA = 128, 10 Hz		20 (5 V) 19.5 (3.3 V)		Bit
Integral nonlinearity	PGA = 128		±15		ppm
Offset error	PGA = 128		±1.4		μV
Offset drift	PGA = 128		20		nV/°C
Gain error	PGA = 128		±0.5		%
Gain error drift	PGA = 128		8		ppm/°C
Reference input					
Reference input	REFIN	1.5	VDD	VDD+0.1	V
Reference output					
Reference output	REFOUT		VDD		V
Clock					
Internal oscillator frequency			5.2		MHz
Internal clock temperature drift			250		ppm/°C
Temperature sensor					
Temperature measurement error	TempError		±3		°C

Table 4 – Power supply characteristics (VDD = 5 V)

Parameter	Condition	Min	Typ	Max	Unit
Supply voltage	VDD	4.5	5	5.5	V
Operating current (normal mode)	PGA = 1, 2		1.57		mA
	PGA = 64, 128		2.34		mA
	Power down		0.1	0.1	μA

Table 5 – Power supply characteristics (VDD = 3.3 V)

Parameter	Condition	Min	Typ	Max	Unit
Supply voltage	VDD	3	3.3	3.6	V
Operating current (normal mode)	PGA = 1, 2		1.26		mA
	PGA = 64, 128		2.11		mA
	Power down		0.1		μA

1.7 Pin Description

Figure 2 – SOP8 pinout: pins 1–4 (left, top to bottom): REFIN, GND, AINN, AINP; pins 5–8 (right, bottom to top): SCLK, DRDY/DOUT, VDD, REFOUT.

Table 6 – Pin description

No.	Pin name	I/O	Description
1	REFIN	AI	Reference input
2	GND	P	Chip ground
3	AINN	AI	Channel negative input
4	AINP	AI	Channel positive input
5	SCLK	DI	SPI clock input
6	DRDY/DOUT	DI/DO	SPI data input / output
7	VDD	P	Power supply
8	REFOUT	AO	Reference output

Note: REFOUT is also the sensor excitation output (output value equals VDD).

2 Functional Block Description

2.1 Analog Input Front End

The CS1237 has one ADC with one differential input. The input signal can be the differential input AINP/AINN or the output of the temperature sensor; switching is controlled by register bits ch_sel[1:0]. Basic structure (Figure 3): temperature sensor, AINP and AINN → input mux → PGA → 2nd-order sigma-delta ADC, with REF as the reference.

The PGA gain (1, 2, 64, 128) is controlled by register bits pga_sel[1:0].

The reference voltage can be supplied externally or generated internally. To use an external reference, the internal reference must first be switched off; this is controlled by register bit refo_off.

2.2 Temperature Sensor

The chip provides temperature measurement. When ch_sel[1:0] = 2'b10, the ADC input is connected to the internal temperature sensor and all other analog inputs are ignored. The ADC derives the actual temperature from the voltage difference measured at the internal sensor. When ch_sel[1:0] = 2'b10 the ADC supports PGA = 1 only.

The temperature sensor requires a single-point calibration. Method: at some temperature A, measure with the temperature sensor to get code value Ya. The temperature at any other point B is then:

Temperature(B) = Yb × (273.15 + A) / Ya − 273.15

A is in degrees Celsius. Ya is the code at point A. Yb is the code at point B.

2.3 Low-Noise PGA

The CS1237 provides a low-noise, low-drift PGA that connects to a bridge sensor's differential output (Figure 4). A front-end anti-EMI filter (R = 450 Ω, C = 18 pF) gives a 20 MHz high-frequency cutoff. The low-noise PGA amplifier provides a gain of 64 through RF1, R1 and RF2, and together with the following switched-capacitor PGA forms the 64 and 128 gains. The gains 1, 2, 64, 128 are configured through pga_sel[1:0]. With PGA = 1 or 2, the ×64 low-noise amplifier is switched off to save power. When the low-noise PGA is used, the input must stay between GND+0.75 V and VDD−0.75 V; outside this range actual performance degrades.

At the CAP pins, an internal 45 pF capacitor together with the internal 2 kΩ resistor RINT forms a low-pass filter for the output of the low-noise PGA. This low-pass filter can also serve as the ADC's anti-aliasing filter.

Figure 4 – PGA structure (text description): AINP and AINN each pass a 450 Ω / 18 pF RC filter into amplifiers A1 and A2, linked by feedback resistors RF1, RF2 and gain resistor R1; each amplifier output goes through RINT to the CAP pins, then to a buffer and the ADC.

The CS1237 has a built-in buffer. With PGA = 1 or 2 the buffer reduces problems caused by the low differential input impedance of the ADC, such as insufficient settling time and large gain error. With PGA = 64 or 128 the buffer reduces the settling error, gain error and code drift caused by the low-pass filter formed by the low-noise PGA through RINT = 2 kΩ and CINT = 0.1 μF.

2.4 Clock Source

The CS1237 uses its built-in oscillator to provide the required system clock; typical frequency 5.2 MHz.

2.5 Reset and Power Down (POR & power down)

At power-up, the built-in power-on-reset circuit generates a reset signal and resets the chip automatically.

When SCLK goes from low to high and stays high for more than 100 μs, the CS1237 enters Power-down mode, with power consumption below 0.1 μA. When SCLK returns to low, the chip resumes normal operation.

When the system returns from Power-down to normal operation, all functional settings are as they were before Power-down; no reconfiguration is needed.

2.6 SPI Serial Communication

The CS1237 uses a 2-wire SPI serial interface; data reception and function configuration are done through SCLK and DRDY/DOUT.

2.6.1 Settling Time

At ADC output rates of 10 Hz or 40 Hz, the digital part needs 3 conversion cycles to satisfy the settling-time requirements of the analog input signal and the filter; at 640 Hz or 1280 Hz it needs 4 conversion cycles.

Figure 5 (settling process 1): after power-up / Power-down recovery / channel switch, the analog part settles within t1, then DRDY stays high for t2 until the first valid data is ready. Figure 6 (settling process 2): after a PGA or data-rate switch, the analog part settles within t3, then DRDY stays high for t2.

Param.	Description	Min	Typ	Max	Unit
t1	Analog settling time after power-up / Power-down recovery / channel switch		2		ms
t3	Analog settling time after PGA switch / data-rate switch		0.8		μs
t2 (DRDY/DOUT held high)	10 / 40 Hz		300 / 75		ms
	640 / 1280 Hz		6.25 / 3.125		ms

2.6.2 ADC Output Data Rate

The output data rate is configured through register bits speed_sel[1:0].

Table 7 – Output rate setting

SPEED_SEL[1:0]	ADC output rate (Hz)
00	10
01	40
10	640
11	1280

2.6.3 Data Format

The CS1237 outputs 24-bit two's-complement data, MSB first. The LSB weight is (0.5·VREF/Gain)/(2^{23}−1). Positive full-scale code is 7FFFFFH, negative full-scale code is 800000H. The table shows ideal output codes for different analog inputs.

Table 8 – Ideal output code vs. input signal (note 1)

Input VIN (AINP−AINN)	Ideal output
≥ +0.5·VREF/Gain	7FFFFFH
(+0.5·VREF/Gain)/(2^{23}−1)	000001H
0	000000H
(−0.5·VREF/Gain)/(2^{23}−1)	FFFFFFH
≤ −0.5·VREF/Gain	800000H

(1) Noise, INL, offset error and gain error are not considered. (The original table lists the last row as "≤ +0.5VREF/Gain", which appears to be a typo for −0.5.)

2.6.4 Data Ready / Data In-Out (DRDY/DOUT)

The DRDY/DOUT pin has 4 functions. **First**, when the output is low, new conversion data is ready. **Second**, as the data output pin: once data is ready, after the 1st rising edge of SCLK, DRDY/DOUT outputs the MSB of the conversion data; on every following SCLK rising edge the data shifts by one bit. After 24 SCLKs all 24 bits have been read; if SCLK is then paused, DRDY/DOUT holds the last data bit until it is pulled high just before the next data is ready. When DRDY/DOUT is pulled low again, new data has been converted and the next read can take place. **Third**, on the 25th and 26th SCLK, it outputs the register status flags. **Fourth**, as the register write/read pin: to configure or read the register, the SPI needs to send 46 SCLKs; the command word entered on DRDY/DOUT determines whether it is a register write or read.

2.6.5 Serial Clock Input (SCLK)

SCLK is a digital input pin. The signal must be clean; glitches or slow rising edges may cause wrong data to be read or the chip to enter a wrong state. SCLK rise and fall times must therefore be below 50 ns.

2.6.6 Data Transmission

The CS1237 converts the analog input continuously. When DRDY/DOUT is pulled low, data is ready to be read; the first SCLK shifts out the MSB, and after 24 SCLKs all 24 bits have been read. If SCLK is paused at that point, DRDY/DOUT holds the last bit until it is pulled high. The 25th and 26th SCLK output a flag showing whether the configuration register has been written: DRDY/DOUT = 1 at the 25th SCLK means the Config register has been written with a new value; the 26th SCLK is a reserved bit for chip expansion and currently always outputs 0. The 27th SCLK pulls DRDY/DOUT high; when it is pulled low again, new data is ready and the next conversion can be read. Basic timing is shown in Figures 7 and 8.

Figure 7 (read timing 1): Data Ready (DRDY/DOUT falls) → t4 → SCLK pulses 1…24, bits 23 (MSB) … 0 (LSB) appear on DRDY/DOUT; t5 = pulse width, t6 = clock-to-new-data delay, t7 = hold time of old data, t8 = data update window, t9 = conversion period. Figure 8 (read timing 2): after 24 data bits, SCLK 25 and 26 output update1 and update2 flags, SCLK 27 forces DRDY/DOUT high until the next data ready.

Table 9 – Read timing

Symbol	Description	Min	Typ	Max	Unit
t4	DRDY/DOUT low to first SCLK rising edge	0			ns

Symbol	Description	Min	Typ	Max	Unit
t5	SCLK high or low pulse width	455			ns
t6	SCLK rising edge to new data valid (propagation delay)	455			ns
t7	SCLK rising edge to old data valid (hold time)	227.5		455	ns
t8	Data update; reading of the old data is not allowed		26.13		μs
t9	Conversion time (1/data rate), 10 Hz		100		ms
	40 Hz		25		ms
	640 Hz		1.5625		ms
	1280 Hz		0.78125		ms

2.6.7 Configuration

The CS1237 can be configured through SCLK and DRDY/DOUT. The configuration timing is shown in Figure 9 (write timing and read timing of the Config register, 46 SCLKs each).

Configuration procedure, after DRDY/DOUT goes from high to low:

1. SCLK 1–24: read ADC data. If you do not need to write or read the register, the remaining steps can be skipped.
2. SCLK 25–26: read the register write-operation status.
3. SCLK 27: DRDY/DOUT output is pulled high.
4. SCLK 28–29: switch DRDY/DOUT to input.
5. SCLK 30–36: input the register write or read command word (MSB first).
6. SCLK 37: switch the direction of DRDY/DOUT (for a register write, DRDY/DOUT is input; for a register read, DRDY/DOUT is output).
7. SCLK 38–45: input the register configuration data, or output it (MSB first).
8. SCLK 46: switch DRDY/DOUT back to output and pull it high. update1/update2 are set or cleared.

2.6.7.1 SPI Command Words

The CS1237 has 2 command words, each 7 bits long:

Table 10 – CS1237 command words

Command	Command byte	Description
Write config register	0x65	Write the Config register
Read config register	0x56	Read the Config register

2.6.7.2 SPI Registers

The CS1237 has one register, Config. Reset value 0x0C.

Bit	B7	B6	B5 : B4	B3 : B2	B1 : B0
Description	Reserved	REF output switch	ADC output rate select	PGA select	Channel select

Table 11 – Config register description

Bits	Name	Description
[7]	-	Reserved by the chip. **Default 0; write 0, never write 1.**
[6]	REFO_OFF	**REF output switch:** REF output is on by default. 1 = REF output off. 0 = REF output normal.
[5:4]	SPEED_SEL	**ADC output rate select:** default 10 Hz. 00 = 10 Hz; 01 = 40 Hz; 10 = 640 Hz; 11 = 1280 Hz
[3:2]	PGA_SEL	**PGA select:** default PGA = 128; in temperature mode PGA_SEL = 00. 00 = 1; 01 = 2; 10 = 64; 11 = 128
[1:0]	CH_SEL	**Channel select:** default channel A. 00 = channel A; 01 = reserved by chip; 10 = temperature; 11 = internal short

Note: the reset value 0x0C = 0b00001100 means REF on, 10 Hz, PGA = 128, channel A.

2.6.8 Power Down Mode

When SCLK goes from low to high and stays high for more than 100 µs, the CS1237 enters Power-down mode, in which all chip circuits are switched off and power consumption is close to 0. When SCLK returns to low, the chip returns to normal operation.

Figure 10 shows the Power-down timing: SCLK held high for t10 → chip powered down → SCLK falls → after t11 DRDY goes high, then low again at Data Ready.

Symbol	Description	Min	Typ	Max
t10	SCLK high hold time	100 µs		
t11	Low-level hold time after SCLK falls	10 µs		

3 Package

The CS1237 is housed in an SOP8 package. Dimensions (mm):

Dim.	Min	Max	Dim.	Min	Max
A	4.95	5.15	C3	0.05	0.20
A1	0.37	0.47	C4	0.20 TYP	
A2	1.27 TYP		D	1.05 TYP	
A3	0.41 TYP		D1	0.40	0.60
B	5.80	6.20	R1	0.07 TYP	
B1	3.80	4.00	R2	0.07 TYP	
B2	5.0 TYP		θ1	17° TYP	
C	1.30	1.50	θ2	13° TYP	
C1	0.55	0.65	θ3	4° TYP	
C2	0.55	0.65	θ4	12° TYP	

Figure 11 (SOP8 package outline drawing) – see the original document for the drawing.